Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

AI engineer with 6+ years building production AI systems in regulated enterprise environments. Designed and built Leasel -- a production-grade multi-agent AI platform operating under federal housing compliance constraints (LIHTC, HUD, Fair Housing Act), where AI governance is a hard requirement: the LLM reasons but deterministic or human logic makes every regulated call. Built AgentProbe, an open-source enterprise evaluation toolkit for LLMs. Currently building AI agents with responsible AI guardrails at Intuit (financial services regulations). Proven experience with agentic workflows (LangGraph, multi-agent collaboration, HITL), RAG pipelines, vector databases, prompt engineering, structured outputs, and CI evaluation harnesses. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). Case study: github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

Agentic AI & Multi-agent Systems | Responsible AI & Guardrails | RAG Pipelines | Human-in-the-Loop (HITL) | LLM Evaluation | Regulated Environments | Prompt Engineering | Structured Outputs | Vector Databases | Python (6+ yrs) | Azure Databricks | Enterprise AI Governance

EXPERIENCE

Intuit

Jun 2026 - Present

Software Engineer - AI · Virginia, USA · Financial Services (Regulated)

- Build **AI agents and agentic workflows** using Anthropic Claude in financial applications, with **responsible AI guardrails**: input validation, sensitive-data filtering, structured output schemas, and fallback logic when model output fails safety checks.
- Design and run **LLM evaluation harnesses** (LLM-as-judge, golden-dataset CI regression) to measure accuracy and catch hallucinations before release in a regulated financial environment.
- Ground LLM responses in enterprise data via **RAG pipelines**, ensuring every answer is traceable to a source record -- an auditability requirement in financial services.
- Build ML pipelines (scikit-learn, XGBoost) with feature engineering, model evaluation, and prediction API deployment on AWS; containerize and ship via GitHub Actions CI/CD with observability.

Entrata

Oct 2020 - Jul 2024

Senior Software Engineer · Enterprise SaaS · Federal Housing Compliance (HUD, LIHTC, Fair Housing Act)

- Built and maintained enterprise software operating under federal compliance requirements (LIHTC, HUD, Fair Housing Act, HAP, Housing Choice Voucher) for 8 years -- the same class of governance constraints present in critical-infrastructure environments.
- Delivered 20+ features end-to-end (Python, Java, JavaScript, React, PostgreSQL) across 120+ Agile sprints; owned reliability and operational quality of deployed services.
- AWS S3 migration with Libsodium encryption for 100% PHI compliance; Docker microservices redesign (50% better scalability); Terraform for infrastructure-as-code.
- Maintained HUD income-limit data pipeline for 7 years -- ~180K records annually, zero compliance violations across every release cycle.
- Built Python/Bash automation pipelines eliminating 80% of manual operational workload; reviewed 300+ pull requests; mentored junior engineers.

Entrata India

Sep 2018 - Oct 2020

Junior Software Engineer · Pune, India

- Built compliance modules and data pipelines (Python, Java, PostgreSQL) for affordable housing; delivered 1.6M-record data reconciliation workflows; led AWS S3 migration.

PROJECTS

Leasel -- Multi-Agent AI Platform with Responsible AI by Design Live on GitHub

Production-grade **enterprise agentic solution** built under federal housing compliance constraints. Core governance rule enforced in code: the LLM plans, reasons, and extracts -- but **every regulated eligibility, compliance, and financial decision routes to a human or deterministic logic**, never the model. 8-agent LangGraph StateGraph (planner, retrieval, reasoning, validation, HITL escalation) with tool execution across document knowledge, live HUD income-limit API, configuration state, and historical ticket data. Hybrid-search RAG over structured and unstructured documents (TF-IDF + semantic reranking, OCR ingestion). Evaluated with LLM-as-judge CI harness: **93.6% precision, 87.9% F1** across 1,240 test cases.

Python · LangGraph · FastAPI · Streamlit · RAG · OCR · SQLite · Docker · CI/CD -- github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe -- Enterprise LLM Evaluation Toolkit Live on GitHub

Open-source **enterprise evaluation platform**: 15-endpoint FastAPI backend, multi-provider LLM routing (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama), 5-metric LLM-as-judge scoring (factual accuracy, hallucination, retrieval quality, completeness, compliance), ChromaDB vector database, and three LangGraph agents. CI regression pipeline with golden-dataset splits alerts when agent performance degrades.

Python · FastAPI · LangGraph · ChromaDB · LangChain · Anthropic · OpenAI · Azure OpenAI · GitHub Actions -- github.com/venkatalakshmiparimala/AgentProbe

AI on FHIR -- Regulated Clinical NLP Integration

LLM-powered NLP system translating natural-language clinical questions into FHIR API calls, with HIPAA-compliant RBAC ensuring data isolation by role. Demonstrates AI deployment within a regulated domain (healthcare) -- ~60% reduction in manual lookup time. Flask, React, Docker, BERT, spaCy.

Python · Flask · React · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs -- github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering -- George Mason University — May 2026

GPA 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2017

TECHNICAL SKILLS

Agentic AI: LangGraph, LangChain, Multi-agent Orchestration, Planners, Tool-calling, Memory, HITL, Responsible AI, Guardrails, Explainability

LLMs & Evaluation: Anthropic Claude, OpenAI, Azure OpenAI, Groq, Ollama, Prompt Engineering, Structured Outputs, LLM-as-Judge, Evaluation Harnesses, CI Regression Testing

RAG & Search: ChromaDB, Pinecone, Weaviate, Elasticsearch, Hybrid Search (TF-IDF + Semantic), Embeddings, Document Ingestion, OCR

ML & NLP: Python, PyTorch, scikit-learn, XGBoost, BERT, spaCy, HuggingFace, Pandas, NumPy, PySpark

Cloud & Data: AWS (S3, EC2, Lambda, RDS, Glue, API Gateway, IAM), Azure (OpenAI, Data Factory, **Databricks**, AKS), GCP (Vertex AI), Docker, Kubernetes, Terraform, GitHub Actions CI/CD

Backend & APIs: FastAPI, Flask, Node.js, Java, Spring Boot, TypeScript, React, PostgreSQL, MySQL, REST APIs, Microservices

Compliance & Governance: AI Safety by Design, Regulated Environment Deployment, HIPAA/PHI, HUD/LIHTC/Fair Housing Act, Audit Trails, Observability (Prometheus, Grafana)